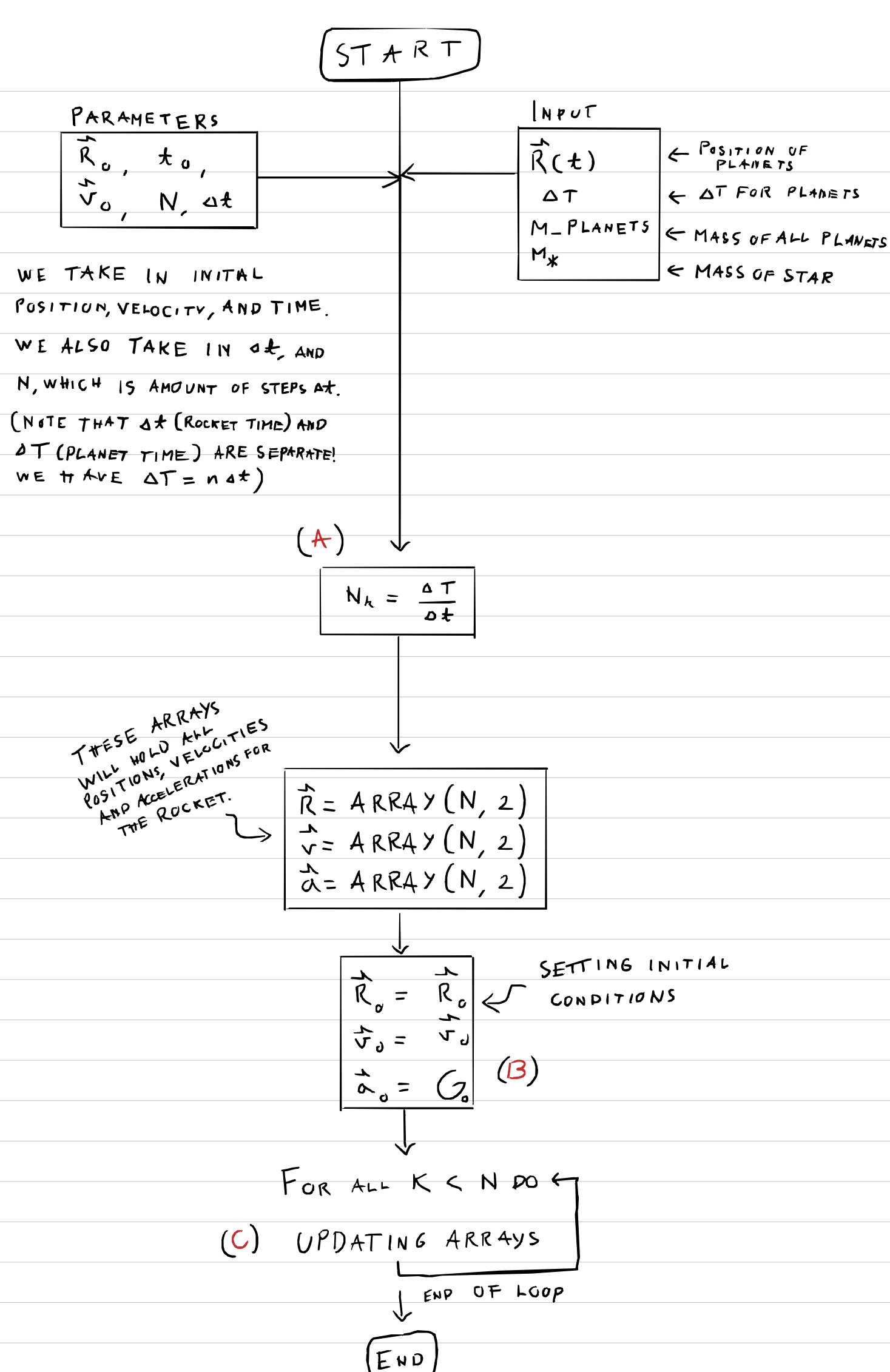
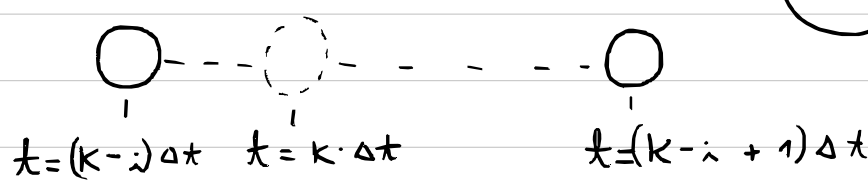




(A)

FOR THIS PROGRAM WE USE A CLOCK STRUCTURE, WHICH HELPS US WITH INTERPOLATION. WE HAVE OUR ITERATIVE VARIABLE K , WHERE $\Delta t \cdot K = T$ (UP UNTIL NOW). SINCE WE ALSO WANT TO KEEP TRACK OF HOW MANY ΔT s HAVE PASSED, WE DEFINE N_K , WHICH IS HOW MANY TIMES Δt GOES INTO ΔT . WE CAN THEN GET THE CLOSEST MULTIPLE OF ΔT , BY TAKING $K - i$, WHERE $K \equiv N_K i$ MODULO.

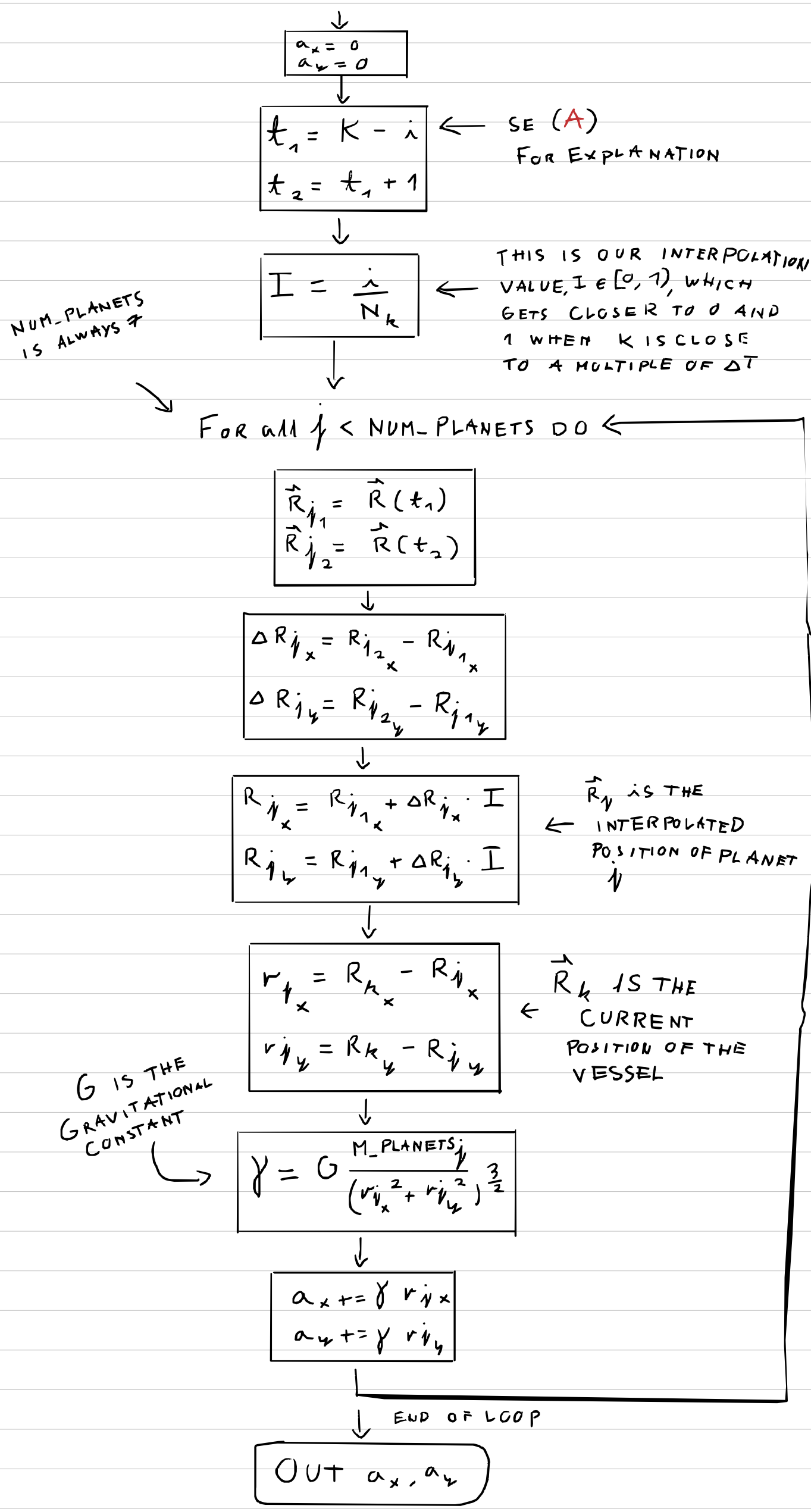


THESE TWO POINTS CAN BE REACHED USING MULTIPLES OF ΔT (PLANET TIME)

THIS ONE CANNOT, AND WE MUST INTERPOLATE

(B)

HERE WE CALCULATE THE GRAVITATIONAL ACCELERATION ON THE VESSEL



(C)

$$R_{k+1,x} = R_{k,x} + v_{k,x} \cdot \Delta t + \frac{1}{2} a_{k,x} \cdot \Delta t^2$$

$$R_{k+1,y} = R_{k,y} + v_{k,y} \cdot \Delta t + \frac{1}{2} a_{k,y} \cdot \Delta t^2$$

$$a_{k+1} = G_{k+1}$$

(B) NOTE THAT INPUT (K WITHIN G) IS NOTED IN SUBSCRIPT

$$v_{k+1,x} = v_{k,x} + \frac{1}{2} (a_{k,x} + a_{k+1,x}) \Delta t$$

$$v_{k+1,y} = v_{k,y} + \frac{1}{2} (a_{k,y} + a_{k+1,y}) \Delta t$$